Lab 2

General System Tasks:

```
michelle@michelleserver:~/Desktop/hello$ apt list --upgradable
Listing... Done
cloud-init/noble-updates 25.3-0ubuntu1~24.04.1 all [upgradable from: 25.2-0ubuntu1~24.04.1]
python3-software-properties/noble-updates 0.99.49.4 all [upgradable from: 0.99.49.3]
software-properties-common/noble-updates 0.99.49.4 all [upgradable from: 0.99.49.3]
systemd-hwe-hwdb/noble-updates 255.1.7 all [upgradable from: 255.1.6]
michelle@michelleserver:~/Desktop/hello$ sudo apt upgrade
[sudo] password for michelle:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages will be upgraded:
  cloud-init python3-software-properties software-properties-common systemd-hwe-hwdb
4 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Need to get 676 kB of archives.
After this operation, 20.5 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 systemd-hwe-hwdb all 255.1.7
 [3,716 B]
Get:2 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 software-properties-common a
ll 0.99.49.4 [14.4 kB]
Get:3 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 python3-software-properties
all 0.99.49.4 [30.0 kB]
Get:4 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 cloud-init all 25.3-0ubuntu1
~24.04.1 [628 kB]
Fetched 676 kB in 2s (408 kB/s)
Preconfiguring packages ...
(Reading database ... 91440 files and directories currently installed.)
Preparing to unpack .../systemd-hwe-hwdb_255.1.7_all.deb ...
Unpacking systemd-hwe-hwdb (255.1.7) over (255.1.6) ...
Preparing to unpack .../software-properties-common_0.99.49.4_all.deb ...
Unpacking software-properties-common (0.99.49.4) over (0.99.49.3) ...
Preparing to unpack .../python3-software-properties_0.99.49.4_all.deb ...
Unpacking python3-software-properties (0.99.49.4) over (0.99.49.3) ...
Preparing to unpack .../cloud-init_25.3-0ubuntu1~24.04.1_all.deb ...
Unpacking cloud-init (25.3-0ubuntu1~24.04.1) over (25.2-0ubuntu1~24.04.1) ...
Setting up cloud-init (25.3-0ubuntu1~24.04.1) ...
Setting up python3-software-properties (0.99.49.4) ...
Setting up systemd-hwe-hwdb (255.1.7) ...
Setting up software-properties-common (0.99.49.4) ...
Processing triggers for udev (255.4-1ubuntu8.12) ...
Processing triggers for rsyslog (8.2312.0-3ubuntu9.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
```

```
michelle@michelleserver:~/Desktop/hello$ sudo reboot

Broadcast message from root@michelleserver on pts/1 (Tue 2026-03-03 00:16:53 UTC):

The system will reboot now!
```

User Tasks:

```
michelle@michelleserver:~/Desktop$ sudo su root
[sudo] password for michelle:
root@michelleserver:/home/michelle/Desktop# sudo useradd bobby
root@michelleserver:/home/michelle/Desktop# ls
hello
root@michelleserver:/home/michelle/Desktop# adduser sally
info: Adding user `sally' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `sally' (1002) ...
info: Adding new user `sally' (1002) with group `sally (1002)' ...
info: Creating home directory `/home/sally' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for sally
Enter the new value, or press ENTER for the default
        Full Name []: Sally
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `sally' to supplemental / extra groups `users' ...
info: Adding user `sally' to group `users' ...
root@michelleserver:/home/michelle/Desktop# sudo su sally
sally@michelleserver:/home/michelle/Desktop$ useradd earl
useradd: Permission denied.
useradd: cannot lock /etc/passwd; try again later.
sally@michelleserver:/home/michelle/Desktop$ exit
exit
root@michelleserver:/home/michelle/Desktop# ls
hello
root@michelleserver:/home/michelle/Desktop# exit
exit
michelle@michelleserver:~/Desktop$ ls
hello
michelle@michelleserver:~/Desktop$ sudo passwd sally
[sudo] password for michelle:
New password:
Retype new password:
```

4. When changing user to root the prompt has root in the front instead of my user name.

5. Useradd doesn't give bobby a home directory or password.

Adduser has a prompt for sally's information. It gives her a home directory.

Adduser utilized the useradd built in linux command.

6. Changing the user to sally changes the prompt to have sally in front. Showing I am acting as sally.

7. She cannot create a new user because she doesn't have sudo commands. Therefore, she doesn't have permission to create a new user.

In this image, I also changed Sally's password, which is task 9.

8.

```
michelle@michelleserver:/etc$ sudo useradd bobby
michelle@michelleserver:/etc$ cat passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:992:992:systemd Resolver:/:/usr/sbin/nologin
pollinate:x:102:1::/var/cache/pollinate:/bin/false
polkitd:x:991:991:User for polkitd:/:/usr/sbin/nologin
syslog:x:103:104::/nonexistent:/usr/sbin/nologin
uuidd:x:104:105::/run/uuidd:/usr/sbin/nologin
tcpdump:x:105:107::/nonexistent:/usr/sbin/nologin
landscape:x:106:108::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
michelle:x:1000:1000:michelle:/home/michelle:/bin/bash
sshd:x:107:65534::/run/sshd:/usr/sbin/nologin
sally:x:1002:1002:Sally,,,:/home/sally:/bin/bash
bobby:x:1003:1003::/home/bobby:/bin/sh
michelle@michelleserver:/etc$ sudo userdel -r bobby
userdel: bobby mail spool (/var/mail/bobby) not found
userdel: bobby home directory (/home/bobby) not found
```

```
michelle:x:1000:1000:michelle:/home/michelle:/bin/bash
sshd:x:107:65534::/run/sshd:/usr/sbin/nologin
sally:x:1002:1002:Sally,,,:/home/sally:/bin/bash
michelle@michelleserver:/etc$
```

Bobby is removed from the passwd file. The first picture is the passwd file with bobby and the second image is after he was removed.

10. It is bad practice to stay logged in as root because of human errors. Such as doing something as root that was a mistake or accidentally giving certain permissions to users that they shouldn't have.

11.

```
michelle@michelleserver:~/Desktop$ id
uid=1000(michelle) gid=1000(michelle) groups=1000(michelle),4(adm),24(cdrom),27(sudo),30(dip),46(p
lugdev),101(lxd)
```

Group Tasks:

12. I don't know how to find what groups Ubuntu belongs to. I tried to do *groups Ubuntu,* but I got an error.

13.

```
michelle@michelleserver:/etc$ sudo usermod -aG sudo sally
michelle@michelleserver:/etc$ sudo su sally
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

sally@michelleserver:/etc$ sudo useradd earl
[sudo] password for sally:
sally@michelleserver:/etc$
```

14&15.

```
michelle@michelleserver:/etc$ sudo groupadd cybersec
michelle@michelleserver:/etc$ sudo usermod -aG cybersec sally
michelle@michelleserver:/etc$ sudo -u sally groups
sally sudo users cybersec
michelle@michelleserver:/etc$
```

16. I checked the groups sally is in by the command *sudo -u sally group* I could also use id instead of group. Or I could change the current user to sally and just input group/id.

Permission and Access Control Lists:

17.

```
michelle@michelleserver:~/Desktop$ mkdir lab1
michelle@michelleserver:~/Desktop$ ls -l lab1
total 0
michelle@michelleserver:~/Desktop$ ls -la lab1
total 8
drwxrwxr-x 2 michelle michelle 4096 Mar  3 02:32 .
drwxrwxr-x 4 michelle michelle 4096 Mar  3 02:32 ..
michelle@michelleserver:~/Desktop$ cd lab1
michelle@michelleserver:~/Desktop/lab1$ nano helloWorld.sh
michelle@michelleserver:~/Desktop/lab1$ ls -l helloWorld.sh
-rw-rw-r-- 1 michelle michelle 32 Mar  3 02:38 helloWorld.sh
michelle@michelleserver:~/Desktop/lab1$ sudo chmod +x helloWorld.sh
michelle@michelleserver:~/Desktop/lab1$ ls -l helloWorld.sh
-rwxrwxr-x 1 michelle michelle 32 Mar  3 02:38 helloWorld.sh
michelle@michelleserver:~/Desktop/lab1$ sudo chmod g+wx
chmod: missing operand after 'g+wx'
Try 'chmod --help' for more information.
michelle@michelleserver:~/Desktop/lab1$ sudo chmod g+wx h
chmod: cannot access 'h': No such file or directory
michelle@michelleserver:~/Desktop/lab1$ sudo chmod g+wx helloWorld.sh
michelle@michelleserver:~/Desktop/lab1$ ls -l helloWorld.sh
-rwxrwxr-x 1 michelle michelle 32 Mar  3 02:38 helloWorld.sh
michelle@michelleserver:~/Desktop/lab1$
```

The directory has read, write, and execute permissions for both the owner and group since I am both the owner and group owner. While others only have read and execute.

18. I wrote *echo "Hello World!"* in the bash script. I accidentally made it executable for everyone. I could've just done *chmod o+x helloWorld.sh,* which would allow it to be executable for others. However, I gave group execute permissions later so it didn't effect much in this case.

19. I changed the group permissions to have write and execute permissions by *sudo chmod g+wx helloWorld.sh* I could have also used the octal representation.

20.

```
michelle@michelleserver:~/Desktop/lab1$ sudo getfacl helloWorld.sh
# file: helloWorld.sh
# owner: michelle
# group: michelle
user::rwx
group::rwx
other::r-x
```

21.

```
michelle@michelleserver:~/Desktop/lab1$ setfacl -m u:sally:rw- helloWorld.sh
michelle@michelleserver:~/Desktop/lab1$ sudo getfacl helloWorld.sh
# file: helloWorld.sh
# owner: michelle
# group: michelle
user::rwx
user:sally:rw-
group::rwx
mask::rwx
other::r-x

michelle@michelleserver:~/Desktop/lab1$ ls -la helloWorld.sh
-rwxrwxr-x+ 1 michelle michelle 32 Mar  3 02:38 helloWorld.sh
```

In the end, the owner and group owner have read, write, and execute permissions, and others have read and execute. Sally just has read and write permissions.